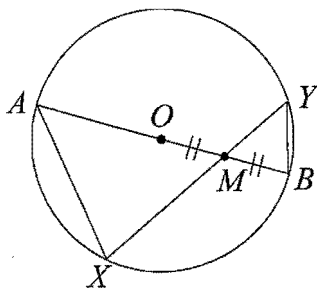


Year 11 Mathematics Extension 1
Assessment Task June 2007

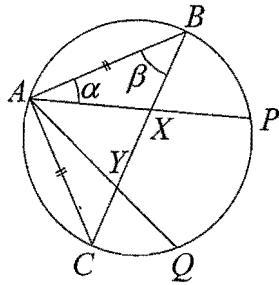
Use a pen. Show all necessary working. Do not use liquid paper. Put one line through your deletions.

- | | | Marks |
|-----|---|-------|
| Q1 | Find all angles θ , for $0^\circ \leq \theta \leq 360^\circ$, for which $\sin 2\theta = \frac{1}{2} \cos \theta$. | 4 |
| Q2 | Given that $\cos 2\theta = 1 - 2 \sin^2 \theta$, show that $\frac{\cos x - \cos(x + 2\theta)}{2 \sin \theta} = \sin(x + \theta)$. | 3 |
| Q3 | a. Show that: $\cos \theta - \sqrt{3} \sin \theta = 2 \cos(\theta + 60^\circ)$. | 3 |
| | b. Hence solve the equation $\cos \theta - \sqrt{3} \sin \theta = 1$ for θ in the interval $0^\circ \leq \theta \leq 360^\circ$. | 2 |
| Q4 | If $\tan(A + B) = \frac{1}{2}$ and $\tan B = \sqrt{2}$, find the value of $\tan A$. | 3 |
| Q5 | a. Show that $\frac{1 + \cos 2A}{\sin 2A} = \cot A$. | 2 |
| | b. Hence find the exact value of $\cot 15^\circ$. | 1 |
| Q6 | $A(-1, 5)$ and $B(5, -4)$ are two points. Find the coordinates of the point P which divides the interval AB internally in the ratio $2 : 1$. | 2 |
| Q7 | A is the point $(-2, 1)$ and B is the point (x, y) . The point $P(13, -9)$ divides AB externally in the ratio $5 : 3$. Find the values of x and y . | 3 |
| Q8 | Find the acute angle between the lines $2x - y = 0$ and $x + 3y = 0$, giving the answer correct to the nearest minute. | 3 |
| Q9 | The acute angle between the line $x - 2y + 3 = 0$ and the line $y = mx$ is 45° . | 2 |
| | a. Show that $\left \frac{2m - 1}{m + 2} \right = 1$. | |
| | b. Find the possible values of m . | 2 |
| Q10 | Two points A and B are on a circle and C is the other end of diameter through A . AE is the line from A perpendicular to the tangent at B . | |
| | a. Draw a careful diagram showing this information. | |
| | b. Prove that AB bisects $\angle CAE$ | 3 |
| Q11 | In the diagram below, AB is a diameter of a circle, whose centre is the point O . The chord XY passes through M , the mid-point of OB . AX and BY are joined. | |



- | | | |
|--|---|---|
| | a. Prove the two triangles formed (triangles AXM and MYB) are similar. | 2 |
| | b. If $XM = 8$ cm and $YM = 6$ cm, find the length of the radius of the circle. | 3 |

Q12

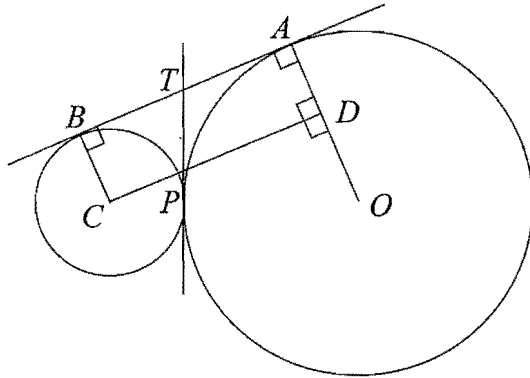


Let $ABPQC$ be a circle such that $AB = AC$, AP meets BC at X , and AQ meets BC at Y , as in the diagram. Let $\angle BAP = \alpha$ and $\angle ABC = \beta$.

- Prove that $\angle BQP = \alpha$.
- Prove that $\angle BQA = \beta$.
- Prove that $PQYX$ is a cyclic quadrilateral.

1
2
3

Q13



Two circles, one with centre C and radius 1 cm and the other with centre O and radius 3 cm, touch each other externally at P . AB is a common tangent to the two circles. CD is drawn perpendicular to OA to complete the rectangle $ABCD$. The common tangent to the two circles at P meets AB at T .

- Find the exact length of AB giving reasons for our answer.
- Find the exact length of PT giving reasons for our answer.

2
2

Total 48

Q1. $\sin 2\theta = \frac{1}{2} \cos \theta$ WR 11 EX 11 JUNE 1972. 240 L - 20 L 25

$$2\sin \theta \cos \theta - \frac{1}{2} \cos \theta = 0 \quad -1$$

$$\cos \theta (2\sin \theta - \frac{1}{2}) = 0$$

$$\cos \theta = 0 \text{ or } \sin \theta = \frac{1}{4} \quad -1$$

$$\therefore \theta = 90^\circ, 270^\circ \quad -1$$

$$\text{or } \theta = 14^\circ 29', 165^\circ 31' \quad -1$$

Q2. LHS = $\frac{\cos x (\cos x \cos 2\theta - \sin x \sin 2\theta)}{2\sin \theta}$ -1

$$= \frac{\cos x (1 - 2\sin^2 \theta) + 2\sin x \sin \theta \cos \theta}{2\sin \theta} \quad -1$$

$$= \frac{\cos x - \cos x + 2\sin^2 \theta \cos x + 2\sin x \sin \theta \cos \theta}{2\sin \theta}$$

$$= \frac{2\sin \theta (\sin \theta \cos x + \cos \theta \sin x)}{2\sin \theta} \quad -1$$

$$= \sin(x + \theta)$$

$$= RHS$$

Q3(a) let $\cos \theta - \sqrt{3} \sin \theta = R \cos(\theta + \alpha)$ -1

$$= R \cos \theta \cos \alpha - R \sin \theta \sin \alpha$$

$$\therefore R \cos \alpha = 1 \quad \therefore \tan \alpha = \sqrt{3} \therefore \alpha = 60^\circ \quad -1$$

$$R \sin \alpha = \sqrt{3} \quad R = \sqrt{1+3} = 2 \quad -1$$

$$\therefore \cos \theta - \sqrt{3} \sin \theta = 2 \cos(\theta + 60^\circ)$$

(b) $\therefore 2 \cos(\theta + 60^\circ) = 1$

$$\cos(\theta + 60^\circ) = \frac{1}{2}$$

$$\theta + 60^\circ = 60^\circ, 300^\circ, 420^\circ \quad -1$$

$$\therefore \theta = 0, 240^\circ, 360^\circ \quad -1$$

Q4 $\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ -1

$$\frac{1}{2} = \frac{T + \sqrt{2}}{1 - T\sqrt{2}}$$

$$1 - T\sqrt{2} = 2T + 2\sqrt{2} \quad -1$$

$$2T + T\sqrt{2} = 1 - 2\sqrt{2}$$

$$T(2 + \sqrt{2}) = 1 - 2\sqrt{2}$$

$$T = \frac{1 - 2\sqrt{2}}{2 + \sqrt{2}} \quad -1$$

Q5. (a) LHS = $\frac{\cos A \sin A + \cos A \sin A + \cos A \sin A}{2\sin A \cos A}$ -1

$$= \frac{2\cos^2 A}{2\sin A \cos A}$$

$$= \frac{\cos A}{\sin A}$$

$$= \cot A$$

$$= RHS$$

(b) $\cot 15^\circ = \frac{1 + \cos 30^\circ}{\sin 30^\circ}$

$$= \frac{1 + \frac{\sqrt{3}}{2}}{\frac{1}{2}}$$

$$= 2 + \sqrt{3}$$

Q6 min = 2:1

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right) \quad -1$$

$$= \left(\frac{2 \times 5 + 1 \times -1}{2+1}, \frac{2 \times -4 + 1 \times 5}{2+1} \right)$$

$$= \left(\frac{9}{3}, -\frac{3}{3} \right)$$

$$= (3, -1) \quad -1$$

Q7. min = 5:-3

$$P = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right)$$

$$\therefore 13 = \frac{5x - 3 \times 2}{2}, -9 = \frac{5y - 3 \times 1}{2} \quad -1$$

$$26 = 5x - 6$$

$$5x = 32$$

$$x = 6.4$$

$$-18 = 5y - 3$$

$$5y = -15$$

$$y = -3 \quad -1$$

Q8. $y = 2x \therefore m_1 = 2$

$$y = -\frac{1}{3}x \therefore m_2 = -\frac{1}{3} \quad -1$$

$$\tan \alpha = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \quad -1$$

$$= \left| \frac{2 - (-\frac{1}{3})}{1 + 2 \times (-\frac{1}{3})} \right|$$

$$= \left| \frac{\frac{7}{3}}{\frac{1}{3}} \right|$$

$$= 7$$

$$\therefore \alpha = 81^\circ 52' \quad -1$$

$$xy(a)2y = x+3$$

$$y = \frac{1}{2}x + \frac{1}{2}$$

$$m_2 = \frac{1}{2}$$

$$\tan 45^\circ = 1$$

$$\tan \alpha = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\therefore 1 = \frac{m - \frac{1}{2}}{1 + m \cdot \frac{1}{2}}$$

$$\therefore \frac{2m-1}{m+2} = 1$$

$$(b) 2m-1 = m+2$$

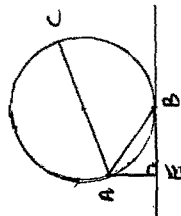
$$m = 3$$

$$\text{or } 2m-1 = -(m+2)$$

$$3m = -1$$

$$m = -\frac{1}{3}$$

Q10.



(b) Draw BC

$$\text{Let } \angle BAE = x^\circ$$

$$\therefore \angle ABE = (90 - x)^\circ \text{ (sum of angles in } \triangle ABE)$$

$$\therefore \angle ACB = (90 - x)^\circ \text{ (both tangents to chord AB are equal)}$$

$$\angle ABC = 90^\circ \text{ (angle in a semi-circle)}$$

$$\therefore \angle CAB = x^\circ \text{ (sum of angles in } \triangle ABC)$$

$$\therefore AB \text{ bisects } \angle CAE \text{ (both } = x^\circ)$$

Q11 (a) In $\triangle AXM$, MYB

$$\angle X = \angle B \text{ (Angles on same arc, stand on same arc are equal)}$$

$$\angle A = \angle Y \text{ (Angles on same arc, stand on same arc are equal)}$$

$$\angle AMX = \angle BMY \text{ (Vert. opp. angles)}$$

$$\therefore \triangle AXM \cong \triangle MYB \text{ (Matching angles equal)}$$

$$(b) AM, MB = XM, MY$$

$$\text{(Prod. of intercepts of chords are equal)}$$

$$= 48$$

$$\therefore (AO + OM) \times OM = 48$$

$$AO = OB = 2 \times OM$$

$$\therefore (2OM + OM) \cdot OM = 48$$

$$3OM^2 = 48$$

$$OM^2 = 16$$

$$OM = 4$$

$$\therefore \text{radius} = 8 \text{ cm}$$

Q12 (a) $\angle BAP = \alpha$ given

$$\therefore \angle BQP = \alpha \text{ (Angles on same arc, stand on same arc are equal)}$$

$$(b) AB = AC \text{ (given)}$$

$$\therefore \angle ABC = \angle ACB \text{ (given)}$$

$$\therefore \angle BQA = \angle B \text{ (Angles on same arc, stand on same arc are equal)}$$

$$(c) \angle AQP = \angle BQP + \angle BQA$$

$$= \alpha + \beta \text{ (from above)}$$

$$\angle AXC = \alpha + \beta \text{ (Ext. angle of } \triangle ABX = \text{sum of int. opp. angles)}$$

$$\therefore PQAX \text{ is cyclic (Ext. angle of cyclic quadrilateral = int. opp. angle)}$$

Q13. (a) C, P, O are collinear

(center & point of contact are collinear)

$$\therefore CD^2 = CO^2 - DO^2 \text{ (Pyth.)}$$

$$= (4+3)^2 - (3-1)^2$$

$$= 4^2 - 2^2$$

$$= 12$$

$$AB = CD = \sqrt{12} = 2\sqrt{3} \text{ cm}$$

$$(b) PT = TB \text{ (Tangents from ext. pt. are equal)}$$

$$PT = TA \text{ (Tangents from ext. pt. are equal)}$$

$$\therefore BT = TA \text{ (Both } = PT)$$

$$= \frac{1}{2} BA = \frac{1}{2} CD = \sqrt{3}$$

$$\text{or } r$$